

Tide retrospective: which observables suffice?

the observation operator on homogeneous polynomials

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Abstract

Fix a nondegenerate minimum with Hessian form H and let $\gamma = N(0, H^{-1})$. For a family $S = (\phi_i)$ of continuous polynomially growing observables, define the observation operator on the degree- k homogeneous polynomials, $\mathcal{T}_S : Q \mapsto (\text{Cov}_\gamma[\phi_i, Q])_i$. For two certified losses with equal jets below k , the family's rescaled moment differences are $o(q^{k-2})$ at every ϕ_i if and only if the degree- k Taylor difference lies in $\ker \mathcal{T}_S$. Hence an injective observation operator recovers the degree- k derivative tensor, the monomial words being one such family, and in $d \geq 2$ a family of n observables has a nonzero kernel direction in every degree $k \geq n$, so no fixed finite family sees every homogeneous perturbation at its first rate. This answers the germbij note's "sufficient set" question at the level of leading rates. Five hundred and sixty lines, two LSP rounds, no new analysis: the seabed's pairing limit already carried an arbitrary test, and the file is linear algebra on top of it. Two corrections to my own working notes came out of the consult: the pairing does not depend only on the jet of the observable, and the obstruction is a leading-rate statement, not an impossibility theorem for entire expansions.

1 Setting

The germbij note recovers the Taylor jet of a loss at a nondegenerate minimum from the asymptotic expansions of posterior expectations; the laplace seabed carries this as the degree-by-degree induction `finite_jet_recovery_of_monomial_rates`, whose data at step k are the d^k monomial words $x \mapsto \prod_j x_{m_j}$. The user's question for this tide was which sub-families of observables suffice, whether there is a clean condition, and what is lost when one takes fewer. The seabed's answer was implicit: the step- k engine `tendsto_pairwise_normalized_moment_difference` states that for *any* continuous polynomially growing test P the rescaled difference $(M_1(P, q) - M_2(P, q))/q^{k-2}$ tends to $-\text{Cov}_\gamma[P, Q_k]$, where Q_k is the degree- k Taylor difference of the two losses. Everything a family can or cannot see at the leading rate is in that pairing. The tide made the statement explicit.

The deliberation (tide log `tide-log/2026-09-19-14-40-tide-germbij-sufficient.md`, consult in `gpt_germbij_sufficient_v1.md`) proposed the sufficiency theorem (A), its converse (B), a finite-family obstruction (C), and an instance construction (B': perturb a certified loss by a kernel polynomial). Both parties voted A+B+C, with B' deferred as a one-line remark for the paper.

2 The thing formalised

Let H_k be the degree- k homogeneous polynomials on $\mathbb{R}^d$ (the span of the monomial words). Let ϕ_i , $i \in \iota$, be continuous with polynomial growth, and let $\mathcal{T}_S(Q) = (\text{Cov}_\gamma[\phi_i, Q])_i$, a linear map $H_k \rightarrow \mathbb{R}^\iota$.

- (B) For certified losses L_1, L_2 with equal jets below $k > 2$: all differences $M_1(\phi_i, q) - M_2(\phi_i, q)$ are $o(q^{k-2})$ iff $\mathcal{T}_S(Q_k) = 0$.
- (A) If $\ker \mathcal{T}_S = 0$, such data identify the degree- k derivative tensor. The monomial family has $\ker \mathcal{T}_S = 0$.
- (C) If $d \geq 2$ and $|\iota| = n \leq k$, then $\ker \mathcal{T}_S \neq 0$; consequently there is a nonzero $Q \in H_k$ such that any two certified losses with equal lower jets and $Q_k = Q$ have all their S -data $o(q^{k-2})$.

The Lean signature of (A):

```
theorem iteratedFDeriv_recovery_of_family_rates (hk : 2 < k)
  (A1 : HigherLaplaceDomain k L1 H) (A2 : HigherLaplaceDomain k L2 H)
  (hlower : forall j < k,
    iteratedFDeriv R j L1 0 = iteratedFDeriv R j L2 0)
  (hsymm1 : (iteratedFDeriv R k L1 0).IsSymm)
  (hsymm2 : (iteratedFDeriv R k L2 0).IsSymm)
  {iota : Type*} (phi : iota -> EuclidD d -> R)
  (hphic : forall i, Continuous (phi i))
  (hphig : forall i, HasPolynomialGrowth (phi i))
  (hinj : forall Q in homogPolySpan d k,
    (forall i, gaussianCovariance H (phi i) Q = 0) -> Q = 0)
  (hdata : forall i,
    (fun q => A1.rescaledMoment (phi i) q - A2.rescaledMoment (phi i) q)
    =o[nhdsWithin 0 (Ioi 0)] fun q => q ^ (k - 2)) :
  iteratedFDeriv R k L1 0 = iteratedFDeriv R k L2 0
```

with

```
homogPolySpan d k := Submodule.span R (Set.range monomialTest).
```

The other declarations:

- the operator itself, `pairingMap`, a linear map on the span;
- (B): `family_rates_iff_pairing_eq_zero`;
- (C): `exists_kernel_of_finite_family` and `finite_family_leading_rate_blind`;
- the monomial instance: `monomialTest_family_injective`.

The hypotheses of (A) and (B) are those of the seabed's step- k theorem; nothing was added. The injectivity hypothesis quantifies over the finite-dimensional span, not over all continuous homogeneous functions, on Astra's advice: the latter space is infinite-dimensional for $d \geq 2$ and no finite family could ever satisfy it, so the theorem would have been true and useless.

3 Proof strategy

(B) is the pairing limit plus uniqueness of limits in one direction, and the pairing limit with target 0 plus `isLittle0_iff_tendsto` in the other. (A) is (B) followed by injectivity, then the diagonal argument copied from `DegreeRecovery`: $Q_k = 0$ gives equal diagonals of the two symmetric tensors, and `iteratedFDeriv_eq_of_diag_eq` polarises. The one new fact is that Q_k lies in the span, which is the seabed’s own `diag_eq_sum_monomialTest` read as a membership statement.

For (C) the span is finite-dimensional (`FiniteDimensional.span_of_finite`) and the $k + 1$ two-coordinate words $x_0^{k-j} x_1^j$ are linearly independent: evaluate any relation along the line $(s, 1, 0, \dots)$ to get a real polynomial in s vanishing identically, kill it with `Polynomial.funext`, and read off each coefficient. So $\dim H_k \geq k + 1 > n = \dim \mathbb{R}^n$ and `LinearMap.ker_ne_bot_of_finrank_lt` produces the kernel element. The two-coordinate count is sharp for $d = 2$; the full count $\binom{k+d-1}{d-1}$ fails earlier in higher dimension and was not needed. Linearity of the covariance in the second slot, which the seabed did not have, is three short lemmas from `integrable_mul_quadKernel_of_polynomialGrowth`.

For the monomial instance, pairings against all words control the self-pairing of any span element (induction on the first slot with `gaussianCovariance_comm`), and the seabed’s rigidity theorem finishes.

4 Roadblocks and resolutions

Two rounds, all mechanical.

Section variables versus explicit arguments. Declaring `variable (hd : 2 <= d)` and also writing `(hd : 2 <= d)` on the definitions double-bound the name, so every later application `sliceWord hd j fed j` into the proposition slot. Dropping the section variable fixed twelve errors at once.

rw with lemmas whose function arguments are lambdas. The rewrite

```
rw [<- gaussianExpectation_add hH (hphic.mul hPc) ...]
```

instantiated the lemma from the continuity proofs, whose statement prints the product as `phi * P`, and then failed to find that pattern. Elaborating the lemma first with explicit `(f := fun x => ...)` and closing with `Eq.trans` and `congr 1`; `funext` avoids the pattern match entirely. Same family as the `Pi.add gotcha` in `CLAUDE.md`.

split_ifs did not split an if inside a coordinate projection. A `by_cases` followed by two `rw [if_pos hi]` did. And on this pin `WithLp` is a structure, so `EuclideanSpace.single_apply` is deprecated for `PiLp.single_apply`, and `simp [Fin.ext_iff]` alone decides the resulting `ifs`.

The consult. One round, before writing, and it changed the statements rather than the proofs. The load-bearing paragraph on my working note’s claim that the pairing depends only on the jet of the observable:

GPT-6 Astra: Already in one dimension, set

$$\phi(x) = x^k - c x^{k+2}, \quad c = \text{Cov}(X^k, X^k) / \text{Cov}(X^{k+2}, X^k).$$

Then ϕ has the same k -jet as x^k , but $\text{Cov}(\phi(X), X^k) = 0$. A genuinely H -uniform sufficient condition is $\mathcal{H}_k \subseteq \text{span}\{\phi_i\} + \mathbb{R}\mathbf{1}$ as functions, not merely as jets.

And on the scope of (C):

GPT-6 Astra: C+B+B' do not prove that no finite family recovers the full jet from its entire expansions. A perturbation invisible at order q^{k-2} may be visible later. For example, with $H = I$, take locally $L_\varepsilon(x) = \frac{1}{2}|x|^2 + \varepsilon x_1^3$, $\phi(x) = x_1^2$. The degree-three covariance vanishes by parity. Nevertheless $M_\varepsilon(q) - M_0(q) = 45\varepsilon^2 q^2 + o(q^2)$.

The coefficient checks by hand ($\mathbb{E}X^8 - \mathbb{E}X^2 \mathbb{E}X^6 = 105 - 15 = 90$, halved). Both corrections are now in the file's module docstring and in the working notes for the paper.

5 What was Mathlib, what was new

Mathlib, all lookup:

- spans: `Submodule.span_induction` (the current motive-with-membership form) and `FiniteDimensional.span_of_finite`;
- dimension: `LinearIndependent.of_comp` and `fintype.card_le_finrank`;
- kernel: `Module.finrank_fin_fun` and `LinearMap.ker_ne_bot_of_finrank_lt`;
- polynomials: `Polynomial.funext`, `finsetSum.coeff`, and `coeff_C_mul_X_pow`;
- counting: `Fin.card_filter_val_lt`;
- asymptotics: `isLittleO_iff_tendsto`'.

Seabed: the pairing limit, the Gaussian rigidity theorem, the diagonal expansion into monomial words, the certificates of the words, the polarisation lemma, the integrability of polynomially growing observables against the quadratic kernel.

New: closure of `HasPolynomialGrowth` and `IsHomogeneousOfDegree` under the module operations, the span `homogPolySpan` with its certificates, membership of the Taylor difference, linearity and symmetry of `gaussianCovariance`, the observation operator, and the six theorems. No new analysis.

6 Lessons

- For `CLAUDE.md`: never combine `variable (h : P)` with an explicit `(h : P)` on a declaration in the same section; and elaborate lambda-argument lemmas with named `(f := ...)` and close by `Eq.trans` rather than `rw`.
- For the tide skill: when a statement is about “which data suffice”, consult before writing on the *quantifier domain* of the condition. The move from all homogeneous functions

to the polynomial span cost nothing in Lean and is the difference between a theorem finite families can meet and one they cannot.

- For the paper: a rate-level identifiability theorem needs its scope stated as such. The working notes had drifted from “degree- D observables give the D -jet” to “and nothing more”, which the seabed does not prove and which the cubic example shows is false at the level of full expansions.

7 Follow-ups

- *Instance (B')*. From a kernel direction Q , the pair $L_0 = \frac{1}{2}H(x, x)$, $L_1 = L_0 + Q$ realises the blindness with certified packages: both are polynomials, so the order- k Taylor remainder of L_1 is zero and the quadratic lower bound survives a shrunken ball. Needs the Taylor coefficients of a polynomial in the `taylorHomogeneousTerm` normalisation.
- *Recovery modulo the invisible subspace*. The data determine the Cov_γ -orthogonal projection of Q_k onto $(\ker \mathcal{T}_S)^\perp$; the seabed’s rigidity theorem makes Cov_γ an inner product on H_k . The cheap Lean form is $\mathcal{T}_S Q = \mathcal{T}_S R \iff Q - R \in \ker \mathcal{T}_S$; the substantive one is a quantitative singular-value bound for stable recovery from noisy leading coefficients.
- *Full-expansion identifiability from a finite family*. Open. The cubic example suggests a finite family may recover the jet from all orders through nonlinear terms in the earlier perturbations; the seabed has the forward all-orders expansions but no machinery for the second-order coefficient in the perturbation.
- *The H -uniform spanning criterion*. $H_k \subseteq \text{span}\{\phi_i\} + \mathbb{R}\mathbf{1}$ as functions implies injectivity for every positive-definite H ; a short corollary of the monomial instance and covariance linearity.

Acknowledgements

GPT-6 Astra corrected the quantifier domain of the sufficiency theorem and the scope of the obstruction, and supplied both counterexamples. The seabed is the `germbij` nondegenerate arc, in particular the monomial-tests tide whose diagonal expansion this file reuses, and the normalized-singular tide of the same day, [2026-09-19-06-05](#), whose retrospective sets the paper context.